DATA STRUCTURES

LAB ASSIGNMENT NO. 08

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‘STACKS’

Q1: Convert Infix → Postfix

1) A + B * (C ^ D - E) ^ (F + G * H) - I

- **C ^ D** → CD^
- **C ^ D - E** → CD^E-
- **G * H** → GH*
- **F + G * H** → FGH*+
- **(C ^ D - E) ^ (F + G * H)** → CD^E-FGH*+^
- **B * (...)** → BCD^E-FGH*+^*
- **A + (...)** → ABCD^E-FGH*+^*+
- **(...) - I** → ABCD^E-FGH*+^*+I-

Postfix: **ABCD^E-FGH*+^*+I-**

2) A + (B * C - (D / E ^ F) * G) * H

- **B * C** → BC*
- **E ^ F** → EF^
- **D / E ^ F** → DEF^/
- **(D / E ^ F) * G** → DEF^/G*
- **B * C - (D / E ^ F) * G** → BC*DEF^/G*-
- **(...) * H** → BC*DEF^/G*-H*
- **A + (...)** → ABC*DEF^/G*-H*+

Postfix: **ABC*DEF^/G*-H*+**

Q2: Convert Infix → Prefix

1) A + (B * (C + (D * E)))

- **D * E** → *DE
- **C + (D * E)** → +C*DE
- **B * (...)** → *B+C*DE
- **A + (...)** → +A*B+C*DE

Prefix: **+A*B+C*DE**

2) A * (B + C * (D - E)) + F / G

- **D - E** → -DE
- **C * (D - E)** → *C-DE
- **B + (...)** → +B*C-DE
- **A * (...)** → *A+B*C-DE
- **F / G** → /FG
- **(...) + (...)** → +*A+B*C-DE/FG

Prefix: **+*A+B*C-DE/FG**

Q3: Evaluate the Following Expressions

1) 56+2*34+- (Postfix)

- **5 + 6** = 11
- **11 * 2** = 22
- **3 + 4** = 7
- **22 - 7** = 15

Answer: 15

2) - * 4 5 + 2 3 (Prefix)

- **4 * 5** = 20
- **2 + 3** = 5
- **20 - 5** = 15

Answer: 15

3) + * 2 3 4 (Prefix)

- **2 * 3** = 6
- **6 + 4** = 10

Answer: 10

4) 93*42/- (Postfix)

- **9 * 3** = 27
- **4 / 2** = 2
- **27 - 2** = 25

Answer: 25

THANK YOU!!